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Consideration of Multiple Rock Failure Mechanisms in Predicting Hole Stability Conditions for the Nahr Umr Shale Layer in Southern Iraq

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Abstract

For the development of Nahr Umr sandstone reservoir, in the lower part of this formation in Ratawi field, South Iraq, the standard architecture involves drilling highly deviated intermediate section and setting the 9 5/8" casing shoe within the middle of this formation. This interval corresponds to the Nahr Umr shale, which is known to be a problematic shale across the Middle East region. To define an optimal MW strategy, we conducted an extensive analysis of instability cases experienced by TotalEnergies and those reported in published literature. The various aspects of Nahr Umr shale have been addressed: in-situ stress orientation and stress regime, physical- mechanical properties, strength classification, anisotropy of the strength. We employed a multi-mechanisms failure model to predict the hole stability condition in terms of the minimum mud weight (MMW), including: shear failure, strength anisotropy, shale lamination, mud pressure filtration and creeping for this shale. Azimuthal effects, i.e well orientation, due to the anisotropy of horizontal stresses appear less significant than those of hole deviation angle because of strength anisotropy of the Nahr Umr shale. The strength anisotropy is estimated to have three times more impact than the hole azimuth. The cumulative effect of these failure mechanisms explains the persistent instability in both vertical and deviated wells, with well azimuth playing a relatively minor role. From a strict geomechanical point of view, drilling the high deviated well in NU shale requires a mud weight from 1.40 ~ 1.45 sg. This mud weight range should also be applied to wells with moderate deviation, i.e., between 30° and 60°, essentially due to the anisotropy of the NU shale. Success in drilling through the Nahr Umr shale requires not only appropriate MW but also effective mud sealing, compatible mud chemistry, and rigorous hole cleaning and back reaming practices.

Introduction

The Nahr Umr (NU) formation, deposited during the Albian period in the Mesopotamian Basin, is a prominent siliciclastic unit extensively developed across the southern Arabian Gulf region. It is recognized as a cap rock shale that seals major hydrocarbon reservoirs such as the Zubair in Iraq and Burgan formation in Kuwait, Shuaiba in UAE. The Ratawi field operated by TotalEnergies situates in South Iraq (cf [Figure 1](#) Ratawi field location map). To produce the lower part of the Nahr Umr sandstone reservoir, TotalEnergies

drilling team proposes a standard well architecture involving setting the 9 5/8' casing shoe in the middle of the Nahr Umr shale, followed by horizontal drains in the NU sandstone reservoir. This requires drilling a highly deviated hole up to 85° through the Nahr Umr shale. As the local authorities regulation is not allowing use of OBM, a high-performance water-based mud (HPWBM) is proposed to drill this shale interval. Nevertheless, challenges remain when drilling highly deviated wells in the Nahr Umr shale as recurrent wellbore stability issues have been reported in South Iraq, and more generally in this region.

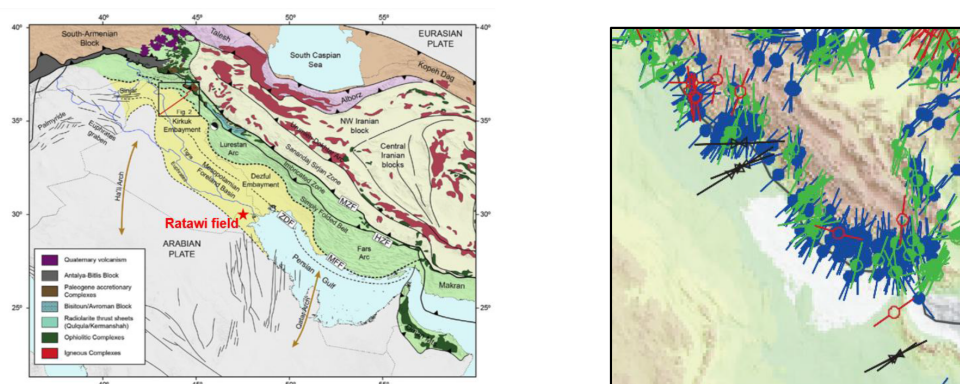


Figure 1—left: Ratawi field location, right-Zoom of world stress map in South Iraq and Gulf Arabics

Indeed, NU formation is geologically characterized by a heterogeneous assemblage of interbedded black shales, fine- to medium-grained quartz arenite sandstones, siltstones, and occasional thin limestone beds. In the location of Ratawi field, the lower part of the NU formation consists of fluvial deposits, which represent the main reservoir of NU interval. Above this, the formation transitions into a deltaic environment, indicating a significant transgressive event. Upsection, the formation becomes increasingly silty and shaly, culminating in an upper interval composed of a mudstone-siltstone-shale mixture, as shown in Figure 2. This reflects the overall basin transgression and the transition toward the overlying Maaddud carbonates. The depositional environment imparts key geology- geomechanics characteristics to the NU shale, defining it as a highly laminated, anisotropic (cf Figure 2, clay-rich unit primarily composed of kaolinite, illite-smectite, and chlorite, and high reactivity to water-based mud.

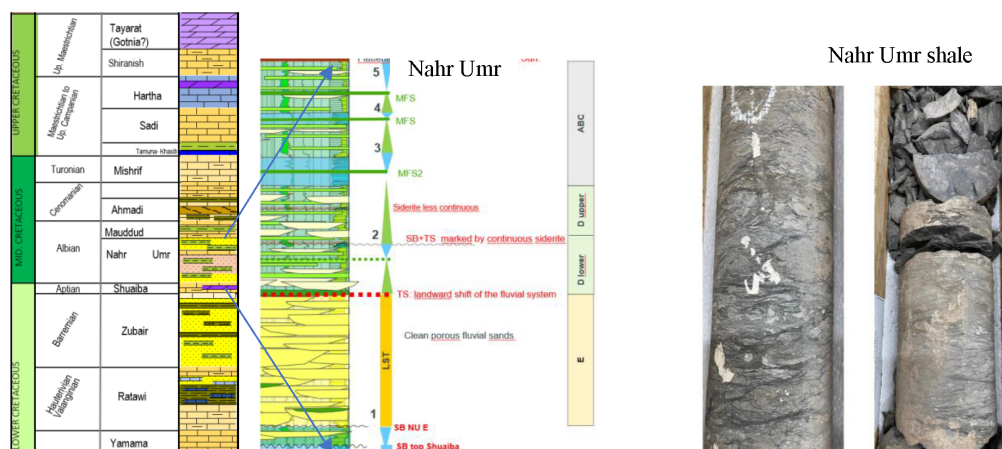


Figure 2—Sequential framework of Nahr Umr (left), photo of shale core of Nahr Umr shale (right), Ratawi field, Iraq

The Nahr Umr shale consists of 75-85% clay minerals, 5-8% quartz, 3-10% calcite, and 7-10% other minerals. The bulk density measured on NU core samples ranges from 2.20~2.5 g/cc. The corresponding dry density is around 1.95-2.25 g/cc, and the porosity lies in the range of 17% ~ 27%. These physical

properties influence the mechanical strength of the Nahr Umr shale, which belongs to the Class 2 - plastic shale, according to the classification of argillaceous rocks given by Hoteit *et al.* 1999, as shown in Figure 3. The typical behavior of such class of shale during a compressive test is characterized by significant phase of plastic strain before reaching peak stress, and a residual strength after the peak stress.

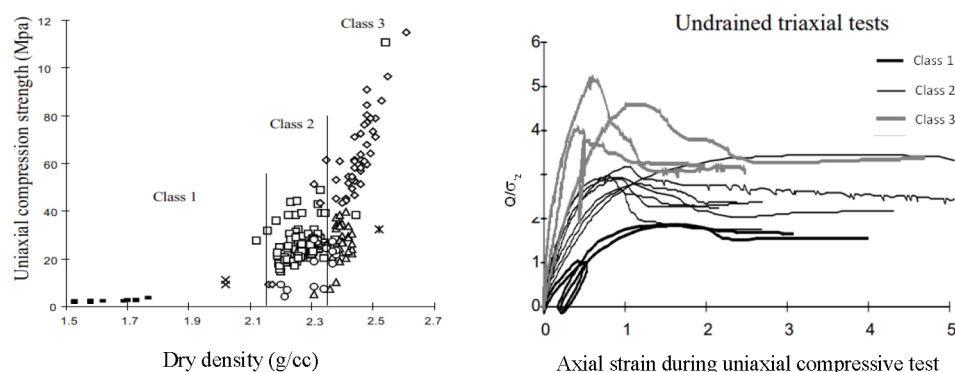


Figure 3—Classification of shale rock mechanics properties (Hoteit *et al.* 1999), Class 1: ductile behavior without strain softening after the deviatoric peak stress, Class 2: plastic shale, between ductile and brittle. Class 3: brittle behavior, failure occurred at low strain ~1%.

Facing the challenge of drilling high deviated well in NU shale, we have carried out an extensive bibliographic analysis of field data, in situ stresses and geomechanical models. We employed a multi-mechanisms failure model to interpret the instabilities recorded and later to predict the hole stability condition in NU shale for drain in Ratawi field. These failure mechanisms are: ①shear failure assuming isotropic Mohr-Coulomb strength model, ②strength anisotropy, ③ shale lamination, ④ mud pressure filtration and ⑤creeping. The cumulative effect of these failure mechanisms explains the persistent instability in both vertical and deviated wells, with well azimuth playing indeed a relatively minor role. Drilling through the Nahr Umr shale demands not only an appropriate mud weight but also effective mud sealing, compatible mud chemistry, and strict adherence to hole cleaning and back reaming procedures.

Field observations of hole instability

Numerous studies have been carried out and published about the hole stability in NU shale. Haider Q.M 2017 has analyzed twenty vertical wells crossed the NU shale in a southern Iraqi oil field. Sever wellbore instability problems have been encountered in 12 ¼" section shale such as stuck pipe, partial and total losses, sidetrack, hole pack off, and caving. Most of these instability issues occurred within the Nahr Umr shale interval. These problems have caused the significant increase of NPT, in some cases, several sidetracks. The author considered the shear failure resulting of the competition of low strength against high stress as the main causes of hole enlargement, stuck pipe and poor primary cement jobs. The analysis reveals that drilled wells with deviation less than 48° is safer and stable.

Al-Enezi, *et al.* 2013 has analyzed the instabilities encountered when drilling through middle Burgan (equivalent Nahr Umr shale) formation. To face the instability in middle Burgan shale and the risk of losses in Mauddud overlaying the NU shale, a change of casing policy is proposed. The casing design was modified and 9%-in casing was set at bottom of Mauddud formation. This enabled to drill the upper Burgan formations with relatively high mud weight. The authors concluded that the effective hole cleaning is essential to prevent tight spots, stuck pipe in NU. High-weighted pills should be pumped based on drilling conditions to ensure proper hole cleaning. High pump rates and minimum 120 RPM pipe rotation are recommended. A dedicated pump-out strategy was elaborated by the authors.

Yu *et al.* (2013) reported complicated instabilities problems experienced in three horizontal wells in the NU formation of Halfaya Oilfield, onshore Iraq. When the first horizontal well using KCL-polymer mud,

there were two sidetracking operations because of pipe stuck, hole collapsing and sticking. The second horizontal well used the organic salt drilling fluid which has a strong inhibition. But in front of the NU shale, hole collapsed and the treatment measure for the sticking failed, sidetrack drilled. The third horizontal well used saturated saltwater drilling, but there were many sticks in NU shale, and cavings at the shaking screen recorded. The importance of mud chemistry to drill NU shale is also highlighted in papers by Nguyen *et al.* 2007 and Yamamoto *et al.* 2002. Despite of low content of reactive smectite and mixed-layered illites / smectites, the NU shale has been observed to have moderate reactivity. These authors underlined that the fractured shale exposed to water-based mud is the most unstable because it absorbs the combined weakening effects of chemical reactivity and fracture flow and deformation.

Yamamoto *et al.* 2004 performed a historical review for the drilling troubles encountered in NU shale based on the data from 1991 to 2004. One of major findings from the review is the effect of mud weight. The chloride content in Biopolymer drilling mud, which has been utilized through the years, increased from 30,000ppm to 150,000ppm for better shale inhibition when the usage of diesel was prohibited in year 1997. Since the number of well sidetracking and tight hole problems increased even after changing the chloride content, mud weight has increased to tackle the shale instability problem. The mud weight has significantly increased since 1997, although the hole angle across NU has remained constant at 35-55 degrees over the years, as shown in Figure 4.

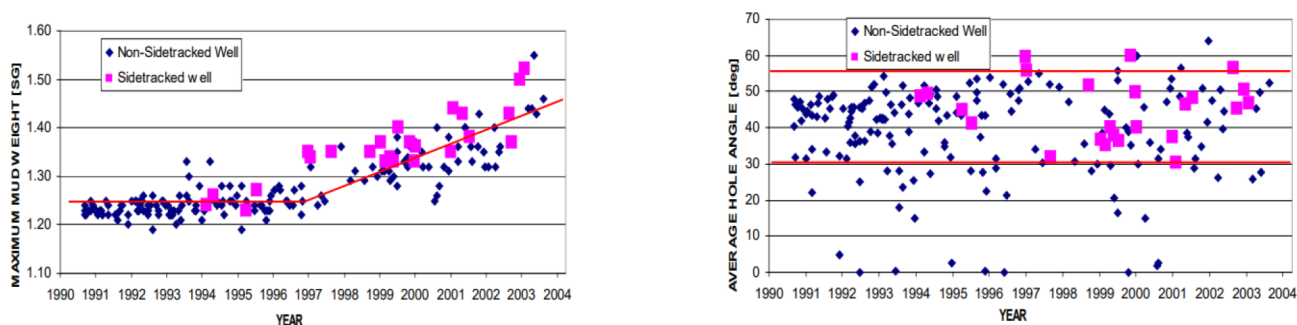


Figure 4—Hole angle and MW used in NU shale (Yamamoto *et al.* 2024)

Based on TotalEnergies' drilling experience through NU shale in offshore wells in the Gulf region, there are no severe hole stability issues for deviations less than 30°, manageable stability for wells deviated between 30° and 45°, and instability that becomes difficult to manage when deviation reaches 45° to 60°. Figure 5 illustrates the mud weight used in 34 wells in front of NU shale, offshore wells. Half of these wells encountered significant challenges, including severe losses during drilling or cementing operations, as well as hole instability issues such as cavings or stuck pipe within NU shale interval. There is no clear correlation between the MW utilized and the problem of instability encountered. The MW applied ranged from approximately 1.30 sg for near-vertical wells to between 1.38 and 1.45 sg for wells deviated at 45°~65°. Hole deviation is restricted to 60° because of systematic instability issues when the inclination is higher than that value. The field experience on NU shale stability shows also no dependence of hole azimuth, and very important role of mud chemistry to avoid sticking and swelling of NU shale.

The cause of the instability issue in NU shale stems from the inherent characteristics of this shale and results from the competition of in-situ stress versus shale rock strength on one side and the alteration of the strength by mud on another side. The conventional method of mud weight prediction, using a single in situ stress and isotropic rock strength model, cannot capture the complexity represented by the NU shale. Moreover, this is not merely a problem of mud weight prediction; it involves also mud chemistry, mud sealing and drilling practice.

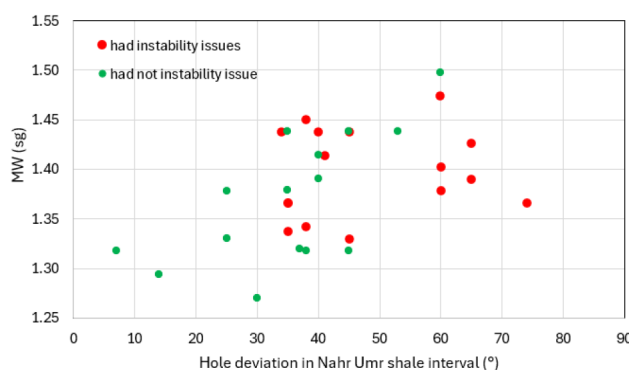


Figure 5—MW used versus hole deviation in NU shale, half of the wells experienced losses and hole instability.

Stress regime in Ratawi field

Knowing the stress regime and stress magnitude is of primary importance in wellbore stability studies. Indeed, they determine not only the mud weight needed to ensure the hole stability, but also the possible optimal well azimuth to avoid the instability issues. Especially in case of a huge contrast of horizontal stresses, drilling in the direction parallel or perpendicular to the maximum horizontal stress (S_{hmax}) can significantly change the stress concentration around the hole and, consequently its stability.

Regarding the azimuth of S_{hmax} in NU shale, there is a consensus that its direction is between N30°E and N55°E in the South Iraq. The range of this orientation is supported by multiple evidence from structural geology analysis, and from the direction of breakout and DIF (drilling induced fracture) observed on hole images, according to the study performed by Jaffar H M, & Abdalnaby W (2018). An example is shown in Figure 6. The prevailing stress direction NE-SW for S_{hmax} is largely attributed to the collision between the Arabian and Iranian plates, which manifests in the formation of the Zagros fold-and-thrust belt. This tectonic interaction, trending NW-SE, results in a high compressive stress in the direction NE-SW, controlling in consequence the regional direction NE-SW of the maximum horizontal stress, as shown in Figure 1-b.

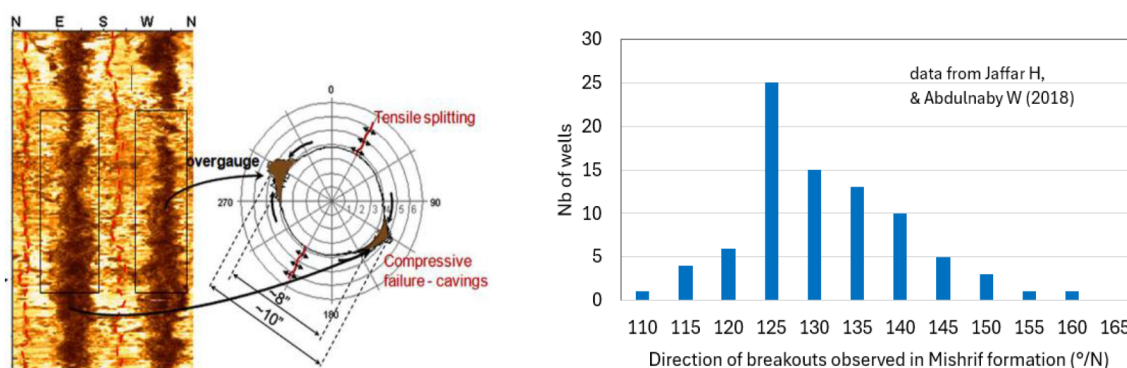


Figure 6—Example of hole image showing the breakouts and drilling induced fracture (DIF) in Mishrif formation Jaffar H, & Abdalnaby W (2018), and distribution of direction of breakouts plotted based on the data reported in this paper

However, regarding the stress regime, there is no clear consensus on whether the investigated field is in a normal faulting or strike-slip stress regime, as different studies report contradictory results. Jaffar H. and Abdalnaby W. (2018) derived a strike-slip stress regime based on borehole image from 84 wells in the Mishrif limestone formation. In contrast, Haider Q.M. *et al.* 2018 proposed a normal faulting regime based on post-mortem analysis of breakouts recorded in onshore wells in South Iraq too. Shaver M.A *et al.* 2021 & 2020 proposed a strike-slip stress model for the NU shale in a field of UAE offshore. The authors consider the NU shale as a vertically transverse isotropic (VTI) material. The S_{hmin} and S_{hmax} profiles have been calibrated using closure and breakdown pressures from microfracture tests. This calibration concludes

a strike-slip regime in the NU shale, while the overlying carbonates remained in a normal faulting regime, a result that appears counterintuitive. Onaisi *et al.* 2000 interpreted also the breakout geometries to determine the S_{hmax} in the ABK field at the reservoir interval suggesting also a marked a strike-slip stress regime. Tsusaka K *et al.* 2021 concluded that the in-situ stress regime in the NU formation can vary from the crest to flank areas. The crest area is the reverse faulting stress regime. The flank area is the normal faulting stress regime at shallower intervals and the strike-slip faulting stress regime at deeper intervals.

As there are non-direct measurements in situ stress available for Ratawi field, we assume a normal stress regime ($S_v > S_{hmax} > S_{hmin}$) in NU shale of Ratawi field with limited contrast between the vertical stress S_v and the stress S_{hmax} as shown in Figure 7. To account for the uncertainty in S_{hmax} , a strike-slip regime ($S_{hmax} > S_v > S_{hmin}$) is also considered for the hole stability sensitivity analysis. Figure 7 shows the depth profiles of in situ stress S_v , S_{hmax} and S_{hmin} elaborated using a combined method of the poroelastic and poroelasto- plastic approach for estimating the S_{hmin} (Su K *et al.* 2022), calibrated with data from vertical wells drilled in the Ratawi field. The S_{hmax} is subsequently estimated by applying a tectonic strain oriented in the northeast direction.

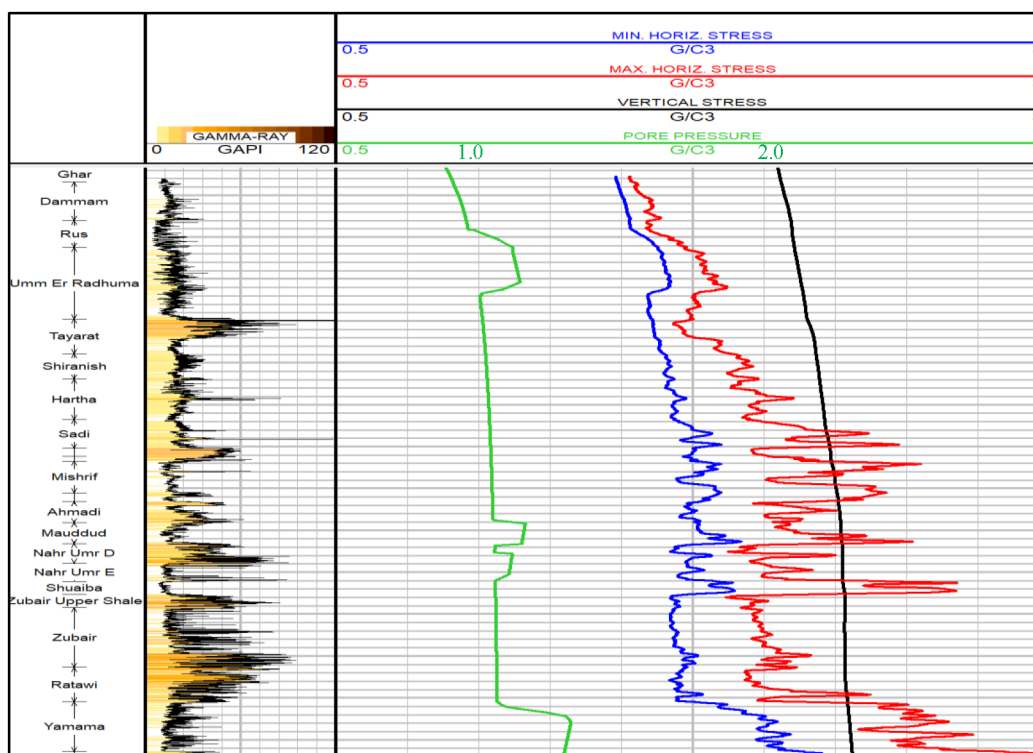


Figure 7—Example of 1D stress model of Ratawi field

Rock strength of Nahr Umr shale and weak bedding

Figure 8 illustrates the stress-strain curves obtained from a multi-stage triaxial compression test conducted by the TotalEnergies Geomechanics Laboratory on a Nahr Umr shale sample. Three confining pressures (P_{conf}): 2 MPa, 5 MPa, and 10 MPa were applied during axial deviatoric loading. The results show a strong dependence of the elastic properties on confining pressure: the Young's modulus increased from 2 GPa at 2 MPa to 5 GPa at 5 MPa, and 7.6 GPa at 10 MPa. To build NU shale strength model, the sonic log of the NU shale facies was used to compute the uniaxial compressive strength, which is estimated in the range 15 and 45 MPa, (cf Figure 9).

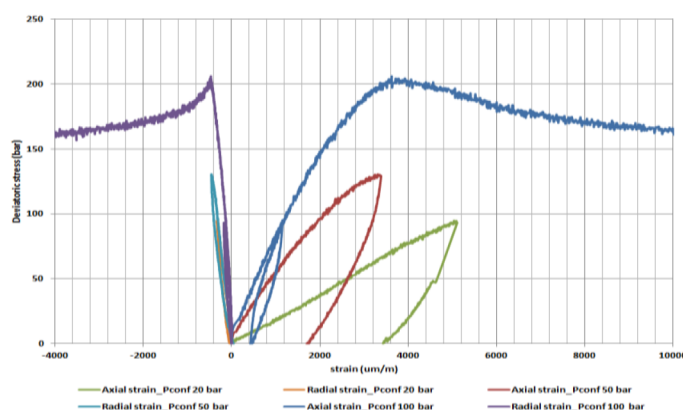


Figure 8—Multistages triaxial compressive test on a Nahr Umr shale sample with 3 confining pressures: 2MPa, 5MPa, and 10 MPa.

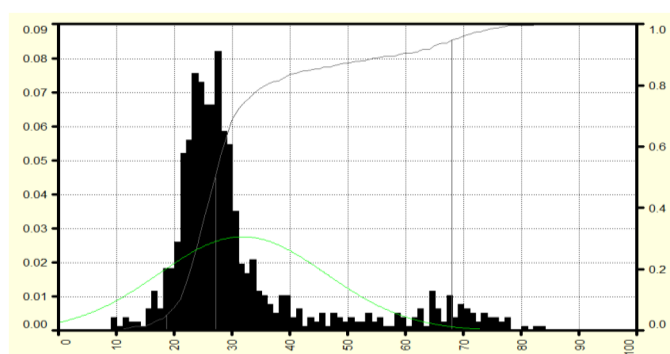


Figure 9—Distribution of UCS of Nahr Umr shale computed from sonic DTc log data of Ratawi field

As for most shales, the NU shale is characterized by abundance of not only bedding planes at meter scale but also network of smaller natural fractures at micro scale (Vinh Nguyen *et al.* 2007). The impact of weak bedding planes on rock strength can be dealt with in two ways: considering the anisotropy of rock strength and taking into account the effect of weak bedding plane.

- **Strength anisotropy:** Figure 10 shows the variation of uniaxial compressive strength (UCS) as a function of the angle between the axial stress and the normal to the bedding plane of a NU shale sample. The lowest strength is observed at an angle of approximately 50° , where the UCS can be up to three times lower than when the stress is applied perpendicular or parallel to the bedding plane.
- **Weak bedding failure:** Hole instability is commonly observed in wells drilled either down-dip or cross-dip relative to the bedding plane of shale (Figure 11). It is believed to be the source of severe instabilities, often characterized by massive caving particularly when the angle between the bedding and the wellbore is low (15° to 20°) as illustrated in Figure 11. The wellbore stability can be very sensitive to this angle, sometimes several degrees can significantly change the hole stability. It is observed that drilling up-dip is much more stable.

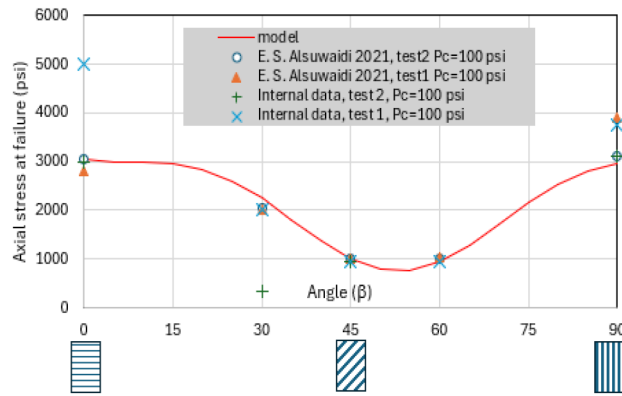


Figure 10—Compressive tests at confining pressure 100 psi carried out on a Nahr Umr shale sample, and the model of anisotropy of compressive strength of Nahr Umr shale (red line)

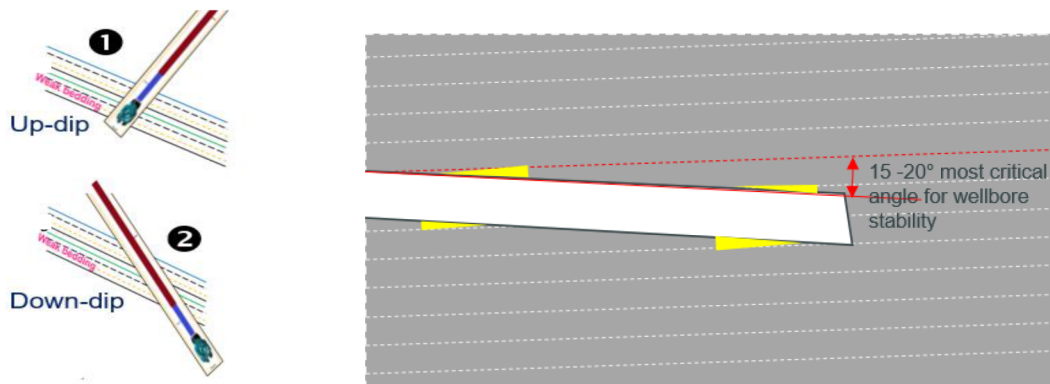


Figure 11—Definition of up-dip and down-dip drilling and illustration of weak bedding effect on hole stability

Simulation of hole stability with superposition of multiple mechanisms of failure

To reproduce the hole instabilities discussed in previous section, and later to predict the stability of horizontal drains in Nahr Umr shale interval in Ratawi field, we have computed the minimum mud weight (MMW) required to address firstly each individual failure mechanism, and later to account for their combined effects. The mechanisms considered are:

1. Shear failure using an isotropic Mohr-Coulomb (MC) strength model for NU shale, $\sigma'_1 = \sigma'_3 \frac{1 + \sin(\phi)}{1 - \sin(\phi)} + \frac{2C \cos(\phi)}{1 - \sin(\phi)}$ where C is the cohesion, and ϕ is the internal friction angle, σ'_1 and σ'_3 are respectively the maximum and the minimum effective stress around the wellbore. We assume the Biot coefficient equal to 1 in the calculation of effective stresses.
2. Shear failure using an anisotropic MC criterion, where the cohesion C varies with hole deviation as shown in Figure 10. It is a pragmatic approximation of the impact of strength anisotropy of NU shale on hole stability.
3. Sliding failure along weak bedding planes: MC criterion in form: $\tau = \sigma'_n \tan(\phi) + C_{wB}$ is used to detect the condition of sliding of weak bedding, C_{wB} is the cohesion of weak bedding plane, which is about half of the cohesion of the intact rock, τ and σ'_n are respectively the shear stress and the normal effective stress on the weak bedding plane around the hole. Elastic stress around a hole for any inclination of the hole can be found in the book Fjaer *et al.* 2008.

The MMW calculations were performed using the TotalEnergies in-house hole stability simulator. The in-situ stress data and mechanical parameters used in the simulations are based on the analysis presented in the previous paragraph and are listed in [Table 1](#).

Table 1—Values of in situ stress and rock strength parameters used in the computation of stereonets of the MMW

Depth	2730 m
Vertical stress S_v	2.26 sg
S_{hmin}	1.70 sg
S_{hmax}	2.05 sg (2.48 sg*)
Pore pressure	1.09 sg
Azimuth Of S_{hmax}	N50°
Cohesion of MC criterion	8.5 MPa
Friction angle	26 °
Dip azimuth of WB	N90 °
Allowable width of breakout	60°
Dip angle of WB	2 °
Cohesion of WB	4 MPa
Friction angle of WB	22 °
Allowable width of breakout on the weak bedding	30°
Anisotropy of strength	Minimum strength at 50° (Figure 10), factor 1/3
*for sensitivity calculation	

[Figure 12](#) presents stereonets of the MMW computed for the above three failure mechanisms. In the top row, the color scale of stereonet is automatically calculated, while in the second row, the color scale is fixed in the range [1.26sg - 1.60sg] to allow assessing the impact of each failure mechanism on the MMW. Based on the comparative analysis of the plots, the following points can be noted:

- Under normal faulting stress regime, and considering isotropic rock strength model, the MMW to ensure hole stability is sensitive to hole deviation ([Figure 12-a](#)). Drilling horizontal well needs 12 points (0.12sg) higher MW than that needed for vertical hole. Drilling in the azimuth of S_{hmin} (N140° or N320°) needs about 3 points sg MW less than that for the well parallel to S_{hmax} (N50° or N230°).
- When an anisotropic strength model is introduced in the simulator of hole stability ([Figure 12-b](#)), the MMW required to drill a well deviated at 40°~ 60° is increased to 1.60 sg, that is much higher than the MW needed for vertical one (1.26sg) and horizontal one (1.38 sg). The result highlights the huge impact of strength anisotropy of NU shale on the wellbore stability and probably explains why the well inclination angle is limited in certain fields in this region. The effect of hole azimuth, i.e drilling parallel to S_{hmax} or to S_{hmin} has not significant effect on the MMW in comparison with the effect induced by the anisotropy of the strength.
- In the 3rd column of [Figure 12](#), the stereonet of MMW to avoid the sliding of quasi horizontal weak bedding combined with the MMW needed to avoid the shear failure of isotropic strength model is plotted. The combined effects, with S_{hmax} oriented at N50 and weak bedding planes, result in a higher mud weight requirement to ensure wellbore stability when drilling in the S_{hmin} direction. The difference in MMW between wells drilled parallel to S_{hmin} and those drilled parallel to S_{hmax} is approximately 0.12 sg.

- In the 4th column of Figure 12, the MMW required to prevent failure due to the three previously mentioned mechanical failure mechanisms is plotted. A quantitative analysis of the AMW data on these stereonet shows that the effect of hole inclination in NU shale is amplified by strength anisotropy. This effect is approximately three times greater than that of well azimuth in consideration of effects of weak bedding.

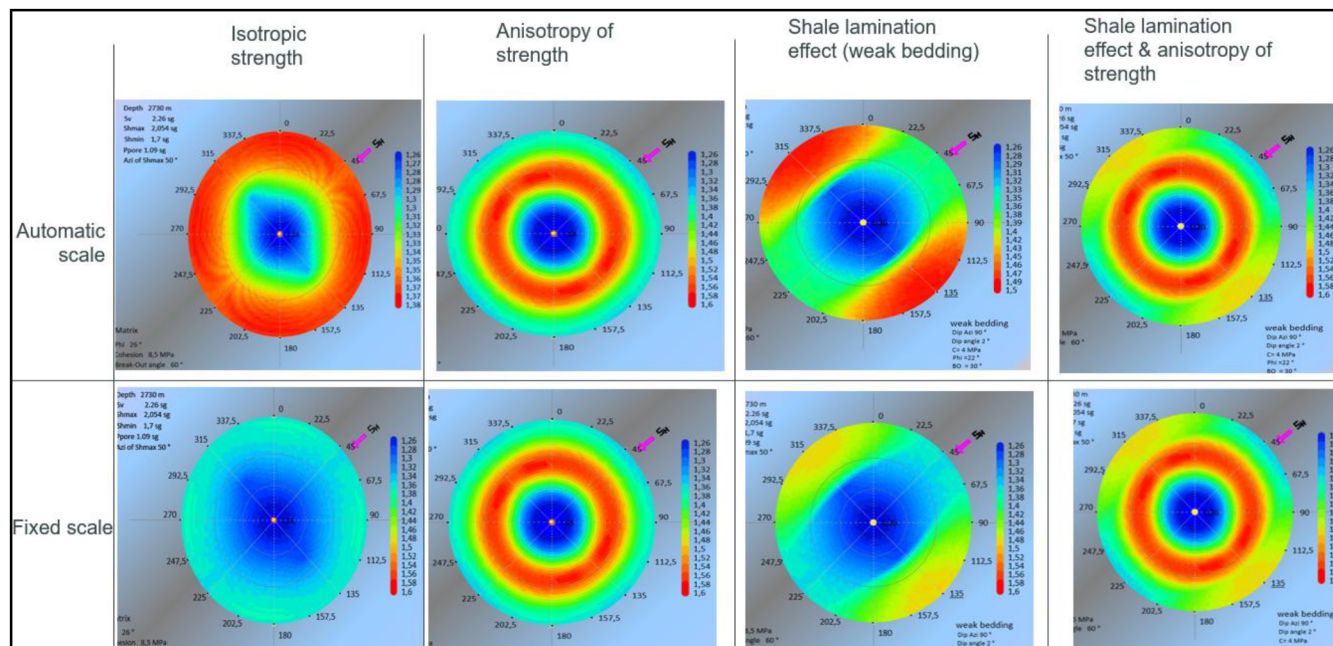


Figure 12-a

Figure 12-b

Figure 12-c

Figure 12-d

Figure 12—Computation of the MMW with superposition effects of anisotropy of strength and weak bedding

Figure 13 portrays the stereonet of the MMW for four scenarios of in situ stresses: normal faulting, strike-slip regime, low contrast or high contrast between S_v and S_{hmax} . The color scale of the plots is fixed in the range [1.26, 1.53] for the sake of comparison. Isotropic MC failure criterion has been applied. We observed that high MW, as high as 1.40~1.48 sg is needed to drill a vertical well in case of strike-slip stress regime, while in case of normal faulting stress regime, the MW needed is 1.26~1.33 sg. The MW used in vertical holes of Ratawi field was ~1.28 sg, despite a certain enlarged hole, no significant instability issue was reported. This back analysis tends to confirm that the strike-slip stress regime in NU shale is not consistent with the stability of vertical hole observed in NU shale interval.

Based on above analysis, from a strict geomechanical point of view, drilling the high deviated well in NU shale interval requires a mud weight from 1.40 ~ 1.45 sg. The hole azimuth effect on the MMW is limited. This mud weight range, even higher, should also be applied to wells with moderate deviation, i.e., between 30° and 60°, essentially due to the anisotropy of the NU shale.

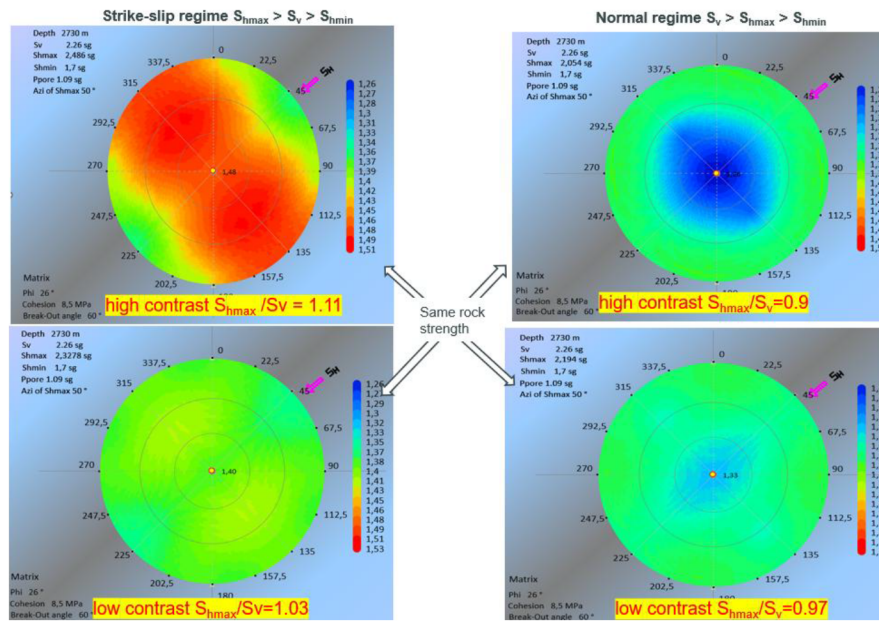


Figure 13—Effects of stress regime and stress contrast between S_{hmax} and S_v on the minimum mud weight (MMW) needed to drill the NU shale interval in Ratawi field

Mud pressure filtration and creeping of shale

All above instability mechanisms can be aggravated by mud filtration which leads to transmission of the mud pressure into the rock around the wellbore, hence a reduced confinement of the rock, or the effect of over balance of mud pressure to maintain the hole stability. Moreover, infiltration weakens the rock especially if the mud is water based. As shown in Figure 14, in Mohr-Coulomb plane, a pore pressure increase in the adjacent rock of a hole pushes the Mohr circle toward the MC shear failure line. When the circle touches the failure line, the rock is doomed to fail, thereby producing cavings, stuck pipe event etc. The extension of failure zone simulated by an elastoplastic model in the simulator is greatly increased (cf Figure 15) due to pressure transmission. This mud infiltration effect is obviously exacerbated by the presence of weak bedding and natural fractures. Adding sealing and inhibitive agents to the mud is therefore crucial for reducing the severity of instability.

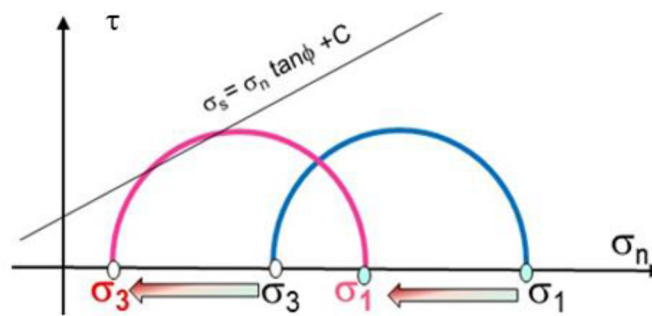


Figure 14—Conceptual model of the impact of mud pressure filtration on hole stability

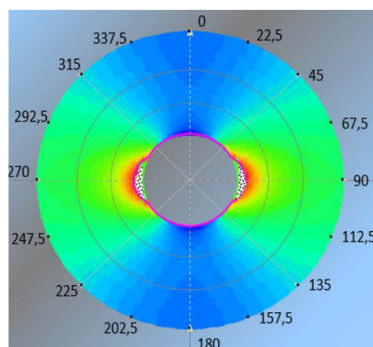


Figure 15-a - without mud pressure filtration

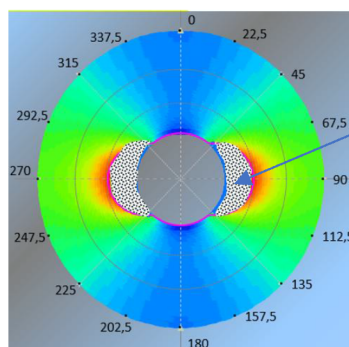


Figure 15-b- with mud pressure filtration

Figure 15—Effects of mud pressure transmission on the extension of plastic zone around a drain in Nahr Umr shale

Poorly consolidated shale, such as Nahr Umr shale, is a typical viscoplastic material. Its ability to creep under deviatoric stress is significant. Due to this creeping behavior, the borehole tends to shrink after drilling, which can lead to stuck pipe incidents when open hole duration is long. Based on laboratory and in situ test data from various shales with a wide range of porosities, SU K *et al.* (2018) proposed a correlation between the rate of creeping and porosity of shale (see Figure 16). For Nahr Umr shale, the creep rate ranges from 10^{-5} to 10^{-6} /h. When cumulative creep deformation reaches 1% to 2%, borehole instability occurs. This can provoke mud invasion in front of fractured shale, weaken the rock strength, and exacerbate instability. To prevent the hole instability due to creeping, in cases of extended open-hole duration, for example more than 10 days, it is preferable to increase the mud weight by 0.01 sg every additional 2 days of open-hole. This rule of thumb is based on field experience acquired in shales with a similar range of porosity.

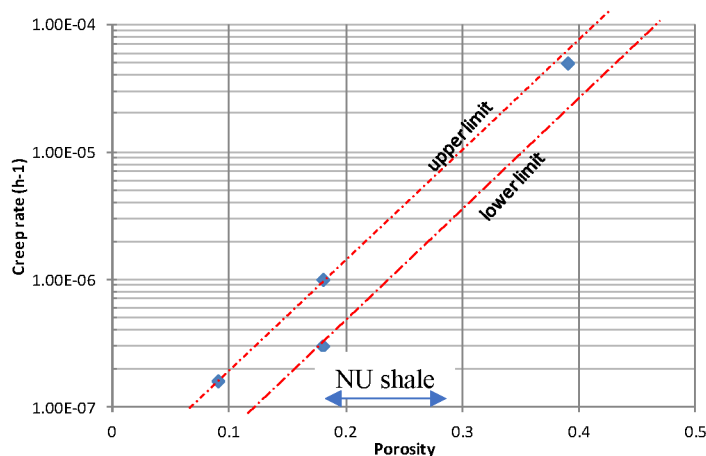


Figure 16—Relationship between creep rate and shale porosity

Conclusions

The Nahr Umr shale is a challenging formation in the Ratawi field and, more broadly, across the Middle East. Despite over 40 years of drilling experience in this formation, it continues to cause significant non-productive time. Operators have encountered instabilities such as cavings, over-gauge, tight spots, pack-offs, and difficulties in back reaming and losses, both in vertical and deviated wells drilled in this shale.

From a geomechanical perspective, the NU shale belongs the class 2, a plastic shale characterized by the presence of natural fractures at various scales and weak bedding planes. Laboratory tests have confirmed the presence of natural fractures and pronounced strength anisotropy of Nahr Umr shale.

There is a consensus on the significant tectonic stresses induced by the collision between the Arabian and Iranian plates. However, the stress regime and the reported contrasts between S_{hmax} and S_v vary across publications and are often inconsistent. In the absence of in situ stress measurements, it is reasonable to assume that the Nahr Umr shale layer at the Ratawi field location is subject to a normal faulting stress regime, with $S_{hmax} - 0.97 * S_v$, indicating a limited stress contrast.

Four mechanisms of failure have been investigated to interpret the recurrent hole instabilities observed in this shale: shear failure, strength anisotropy, weak bedding and mud pressure filtration. The cumulative effect of these failure mechanisms explains the persistent instability in both vertical and deviated wells, with well azimuth playing a relatively minor role. Due to strength anisotropy of this shale, the effect of hole inclination is approximately three times greater than that of well azimuth. With this rationale, there are not enough arguments to conclude whether drilling in S_{hmax} direction is better or worse.

From a strict geomechanical point of view, drilling the high deviated well in NU shale requires a mud weight of about 1.40 ~ 1.45 sg. This mud weight range should also be applied to wells with moderate deviation, i.e., between 30° and 60°, essentially due to strength anisotropy.

We also note that the creep behavior of Nahr Umr shale can be significant. In cases of prolonged open-hole duration, for example, longer than 10 days, it is preferable to increase the MW by 0.01 sg every additional 2 days, based on field experience acquired in shales with a similar range of porosity.

To successfully drill highly deviated wells in the NU shale in Ratawi field, it is essential to combine appropriate mud weight with good mud sealing properties, suitable mud chemistry, and good practices for hole cleaning and back-reaming. Having sufficient mud weight alone is not sufficient to ensure hole stability in this formation.

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